Monitoring and Observability Plan

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1. Monitoring Tools

- Prometheus with the blackbox exporter for metrics and uptime probes
- Grafana for dashboards
- Loki and Promtail for logs
- Alertmanager for alert routing to the on-call pager (Opsgenie) and the Slack channel #jm-alerts
- Sentry for front-end and back-end exceptions

2. Key Metrics

- API request rate, error rate (5xx) and 95th percentile latency
- Matching job duration and queue depth
- Partner synchronisation lag in minutes
- Database connection pool usage and replication lag
- Node CPU, memory and disk usage
- Backup job success

3. Alerting

Alert	Condition	Severity	Response action	Notification channel
HighApiErrorRate	More than 2 percent of API requests return 5xx for 5 minutes	Critical	Follow runbook RB-01; roll back the latest release if it started after a deployment	Pager (Opsgenie) to the on-call engineer
DatabaseConnectionsExhausted	Connection pool above 90 percent for 5 minutes	Critical	Follow runbook RB-02; find and stop the runaway query	Pager (Opsgenie) to the on-call engineer

Alert	Condition	Severity	Response action	Notification channel
MatchQueueBacklog	More than 5000 messages waiting in the matching queue for 10 minutes	Warning	Follow runbook RB-03; scale the Matching Engine workers	Slack #jm-alerts
SyncLagHigh	Partner synchronisation lag above 30 minutes	Warning	Follow runbook RB-04; run the partner sync backfill procedure	Slack #jm-alerts
BackupFailed	A nightly backup job did not finish successfully	Critical	Follow runbook RB-05; rerun the job and check storage	Pager (Opsgenie) to the on-call engineer
CertificateExpiry	A TLS certificate expires in less than 14 days	Warning	Rotate the certificate with the SOP Rotate TLS Certificates	Slack #jm-alerts
DiskUsageHigh	A node or volume is more than 85 percent full	Warning	Clean up or extend the volume	Slack #jm-alerts

4. Dashboards

- API Overview: traffic, errors and latency per route
- Matching Pipeline: queue depth, job duration, model version
- Partner Sync: lag and failures per partner
- Infrastructure: nodes, volumes, databases

5. Log Locations

Application logs are written as JSON to standard output and collected by Promtail into Loki (retention 30 days). Audit logs are written to the audit schema of the primary database and archived monthly to object storage (retention 7 years). Ingress access logs are kept on the ingress nodes under /var/log/nginx for 14 days.